

Kundai Farai Sachikonye

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🌐 github.com/fullscreen-triangle



Computational scientist for pharma: computational pharmacology (PK/PD/ADME), drug metabolism, mass-spectrometry bioanalytics, and precision-medicine data.

Profile

Pharmaceutical biotechnologist and computational scientist working across computational pharmacology (PK/PD/ADME), drug metabolism and mass-spectrometry bioanalytics, biomarker discovery, and multi-omics for precision medicine.

Technical Skills

Computational pharmacology	PK / PD / ADME modelling; binding affinity (K_d); dose-response (Hill); therapeutic-effect prediction
Drug metabolism & bioanalytics	Cytochrome-P450 (CYP) metabolism; LC-MS/MS and high-resolution mass spectrometry; metabolite identification
Precision medicine	Multi-omics integration; biomarker discovery; oncology (neoantigen / MHC)
ML & data	PyTorch, scikit-learn; statistics; reproducible pipelines (Python, R)
Validation	Against reference data — NIST, IEDB, 1000 Genomes / gnomAD / UK Biobank

Experience

2021–present	Independent Computational Pharmacology & Bioanalytics Researcher , <i>Bitspark GmbH</i> Computational pharmacology and mass-spectrometry bioanalytics tools; authored a monograph on cytochrome-P450 drug metabolism; deployed, reference-validated tools.
2019–2021	Computational Lipidomics & Mass Spectrometry , <i>Technische Universität München</i> , Munich High-throughput MS analysis at HPC scale. Taught at Master's level.
2017–2018	Clinical Glycomics & Proteomics , <i>Universitätsklinikum Hamburg-Eppendorf</i> , Hamburg LC-MS/MS and MALDI-TOF pipelines for clinical samples.
2013	Cancer Biomarker Discovery , <i>University Medical Center Groningen</i> , Groningen UPLC-MS/MS shotgun proteomics on tumour samples.

Selected Projects & Publications

Crown Prince	Geometric computational pharmacology — PK/PD/ADME and therapeutic trajectories, binding affinity from first principles; 39/39 NIST validation. crown-prince-four-corners.vercel.app
CYP450 monograph	Published monograph on cytochrome-P450 drug metabolism (2026).

- lavoisier High-throughput mass-spectrometry pipeline for metabolite identification; 98.9% NIST accuracy.
lavoisier.vercel.app/sandbox
- gospel Multi-omics integration for precision medicine; validated against 1000 Genomes, gnomAD,
UK Biobank. github.com/fullscreen-triangle/gospel

Education

- 2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
- 2018–2019 **MSc Computational Biology**, *Universität Tübingen*
- 2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

- English Native / Fluent (C2)
- German Conversational

resident in Germany since 2011